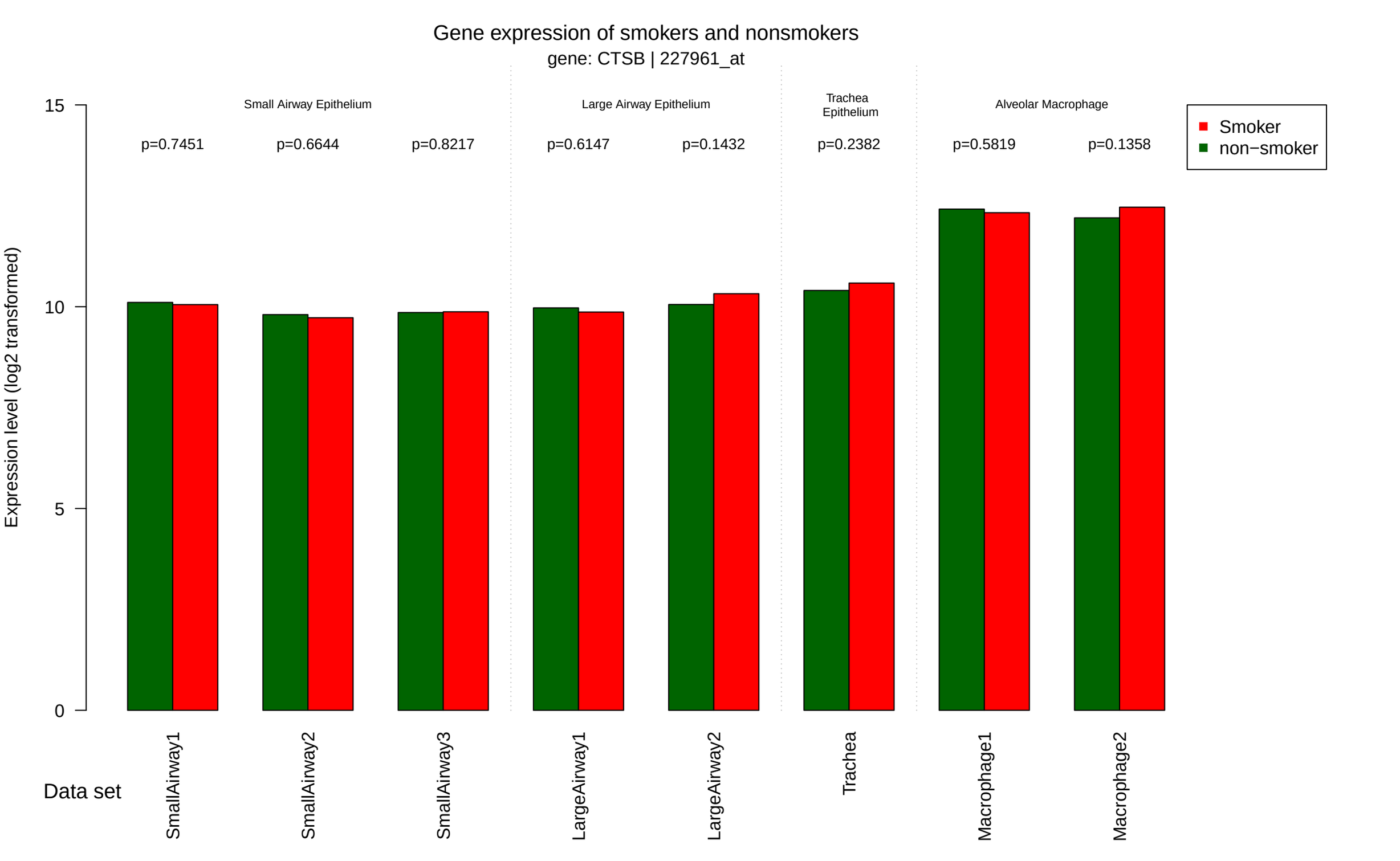
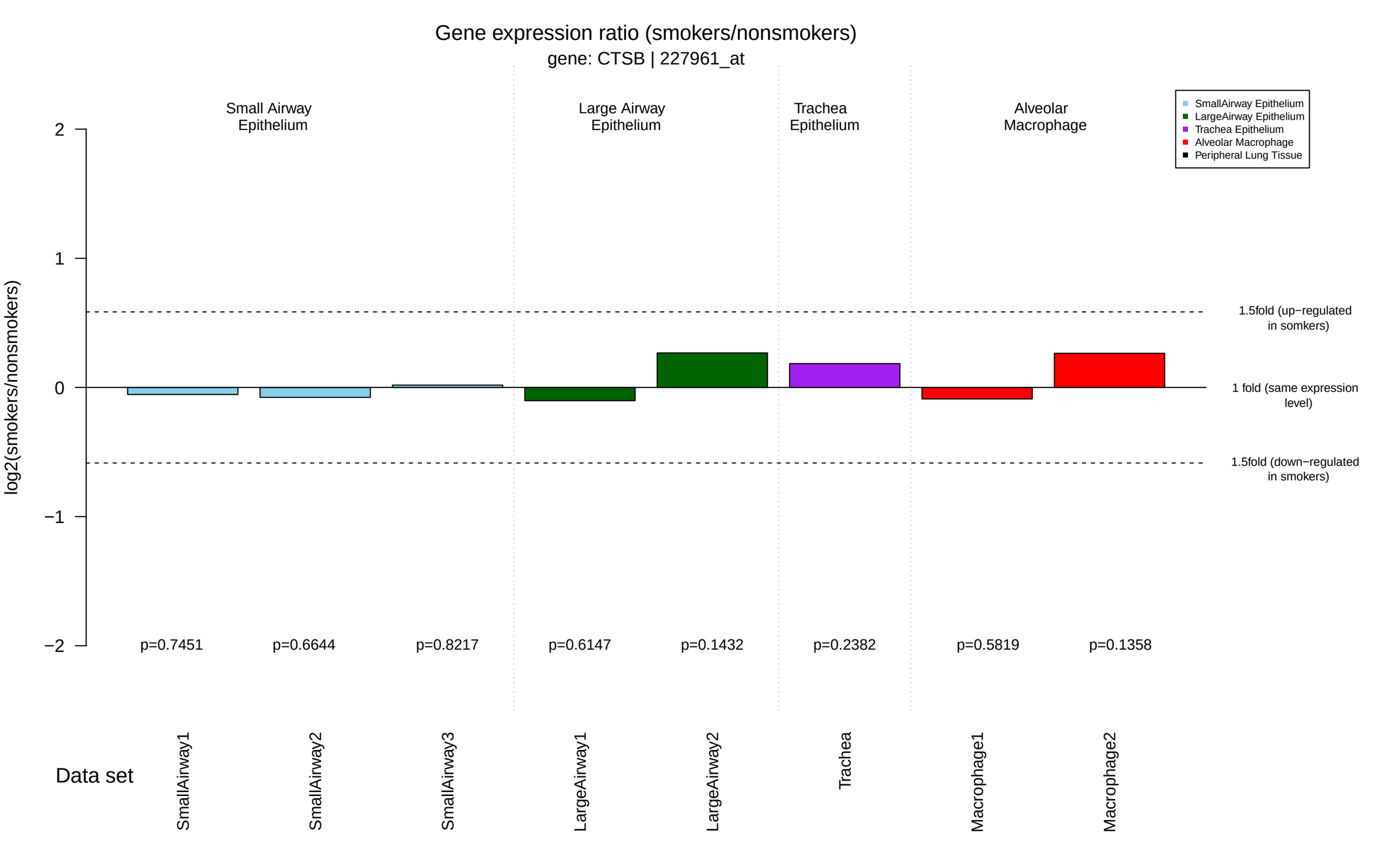
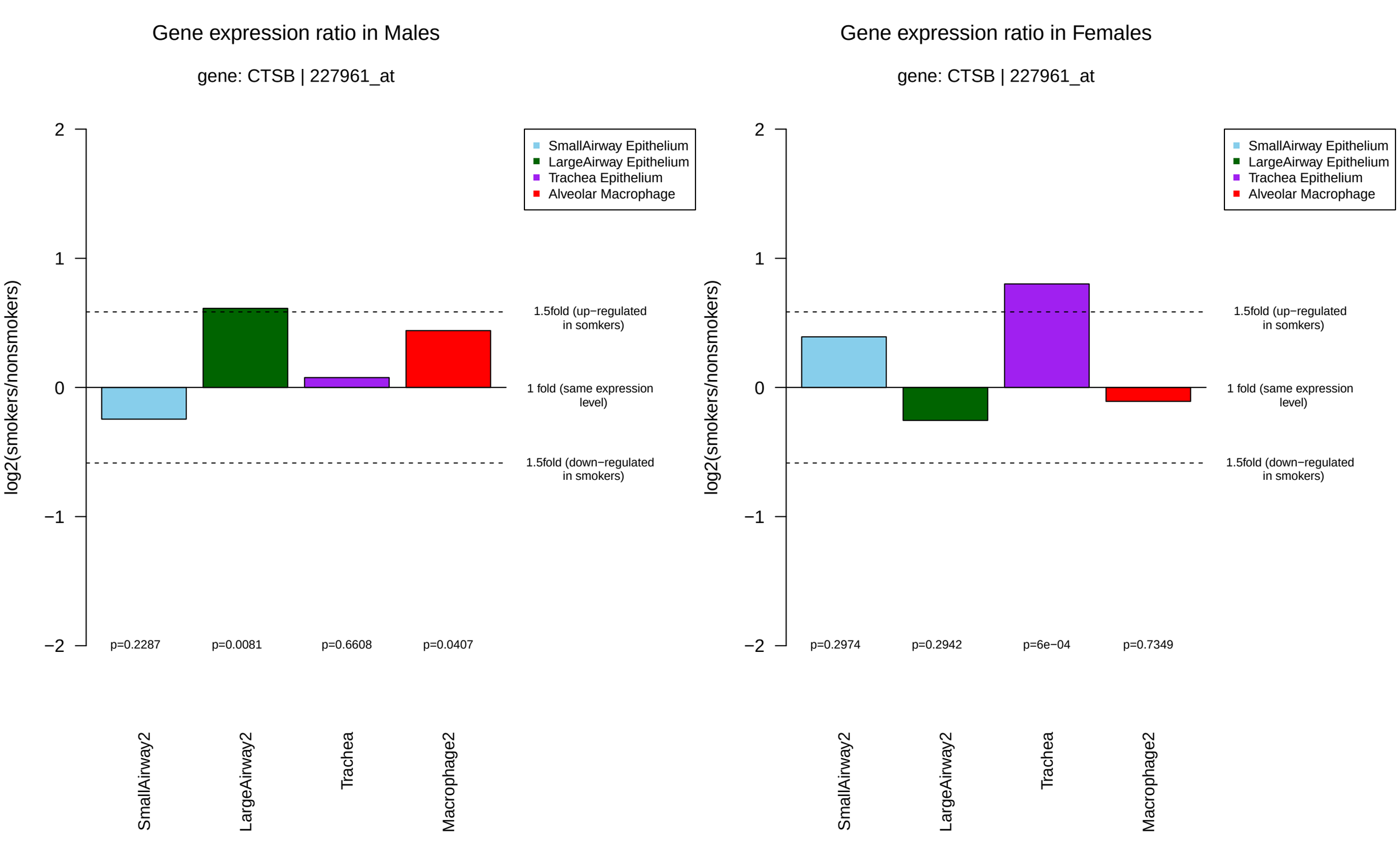
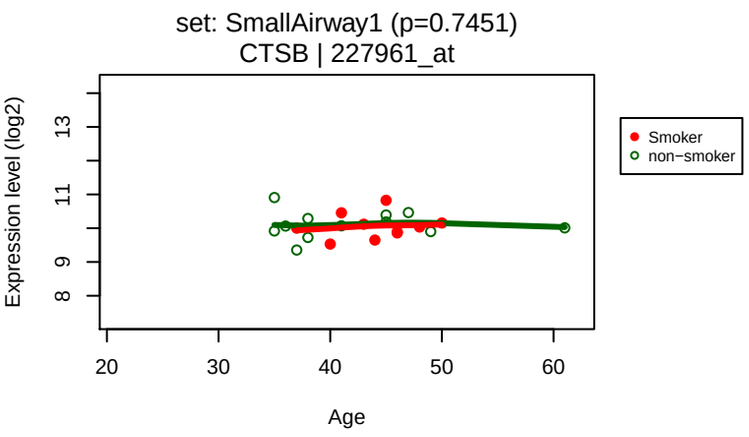
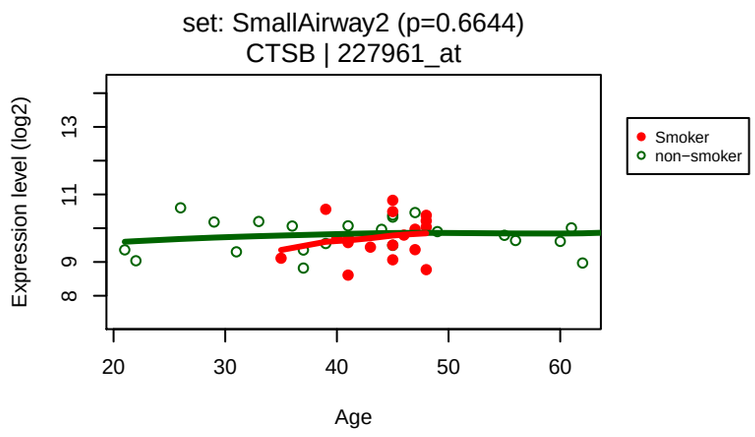
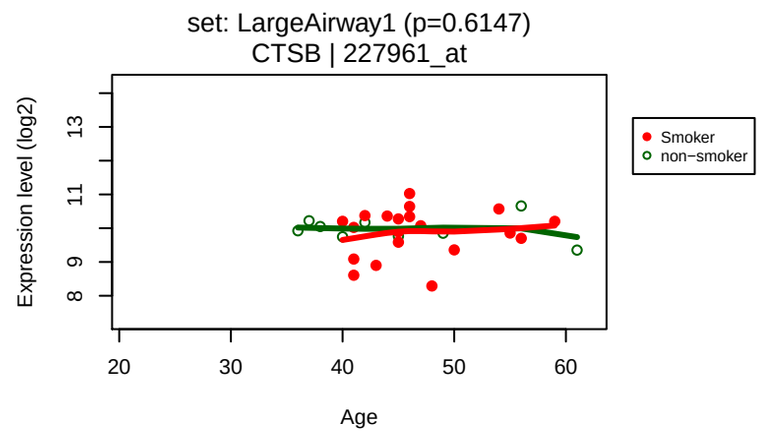

Gene expression of smokers and nonsmokers
gene: CTSB | 227961_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7451
p=0.6644
p=0.8217
p=0.6147
p=0.1432
p=0.2382
p=0.5819
p=0.1358
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

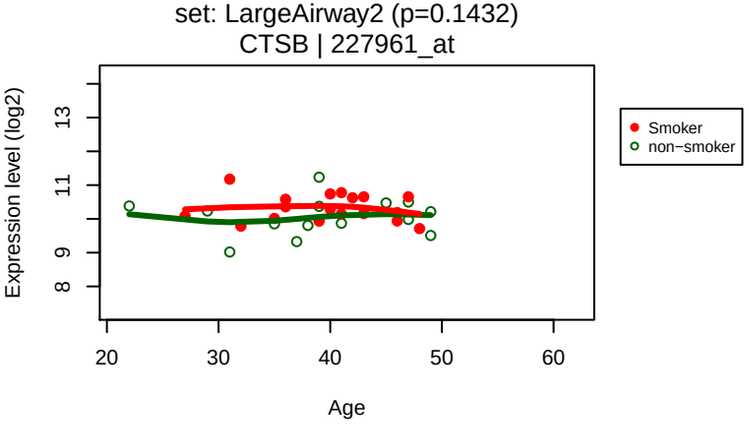

Gene expression ratio (smokers/nonsmokers)
gene: CTSB | 227961_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7451
p=0.6644
p=0.8217
p=0.6147
p=0.1432
p=0.2382
p=0.5819
p=0.1358
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

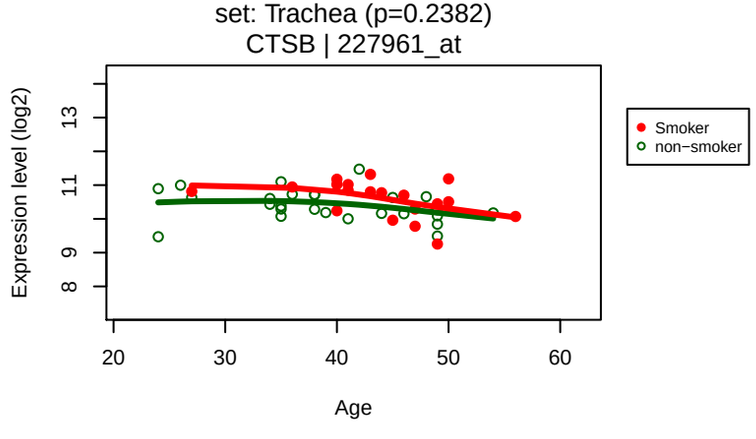

Gene expression ratio in Males
gene: CTSB | 227961_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2287
p=0.0081
p=0.6608
p=0.0407
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSB | 227961_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2974
p=0.2942
p=6e−04
p=0.7349
SmallAirway2
LargeAirway2
Trachea
Macrophage2

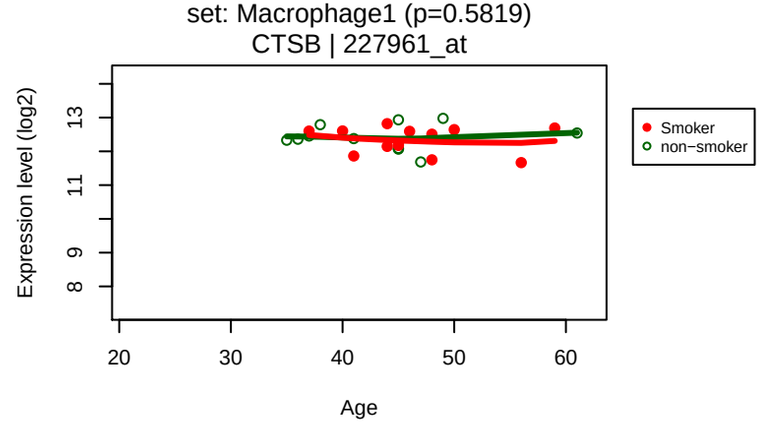

set: SmallAirway1 (p=0.7451)
CTSB | 227961_at
Expression level (log2)
Age
Smoker
non−smoker

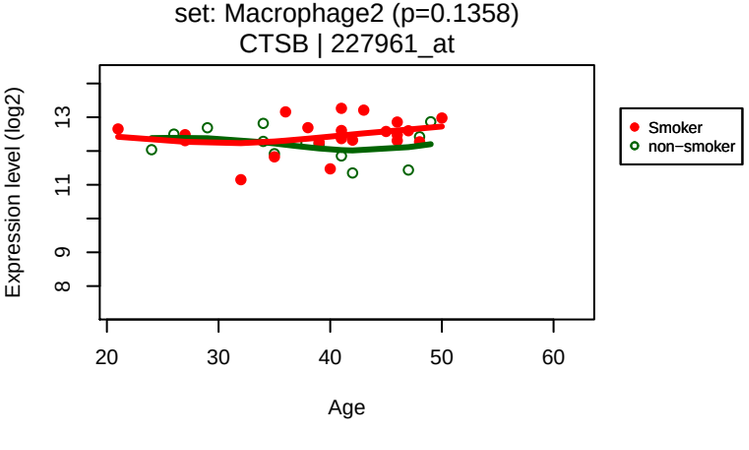

set: SmallAirway2 (p=0.6644)
CTSB | 227961_at
Expression level (log2)
Age
Smoker
non−smoker

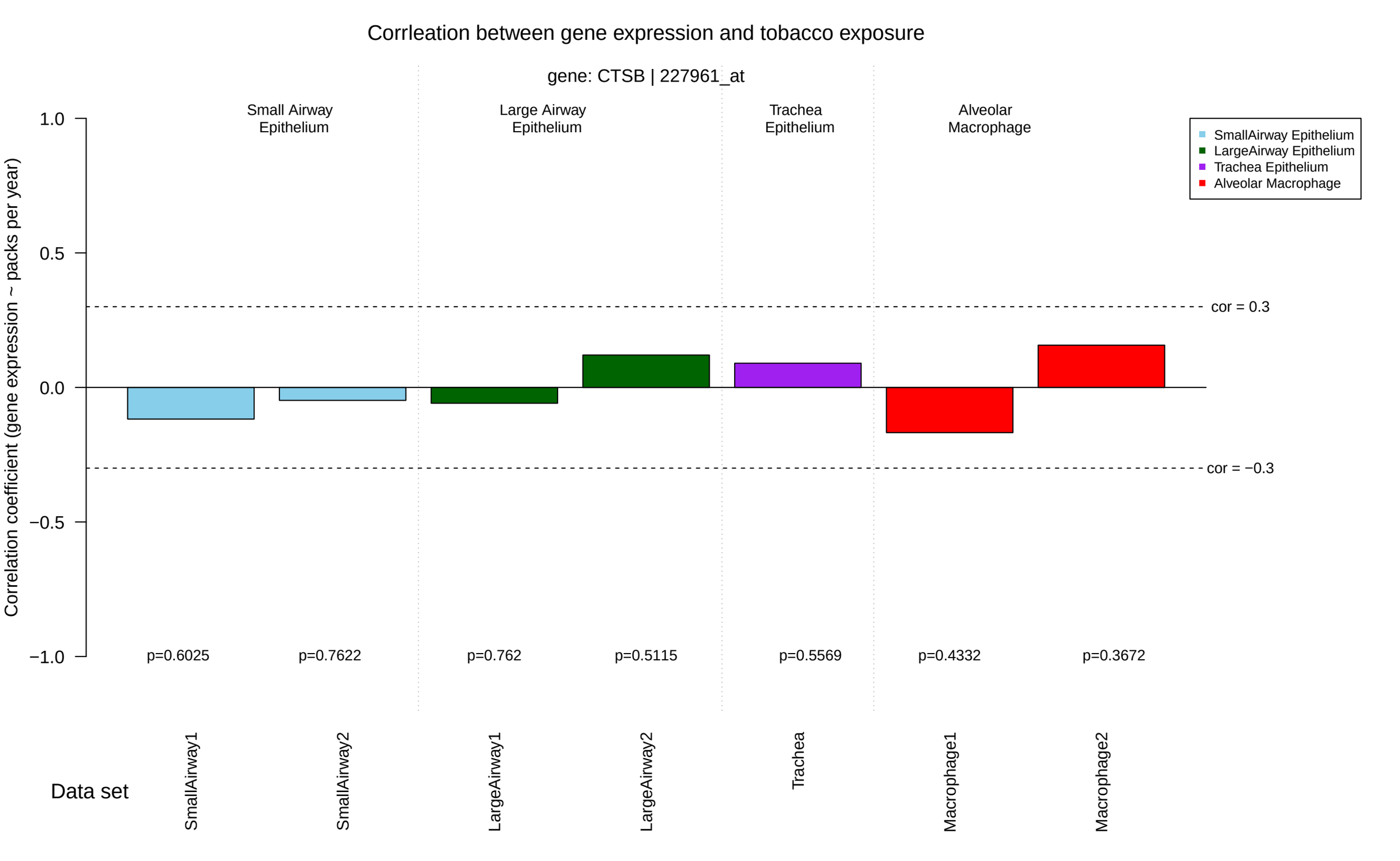

set: LargeAirway1 (p=0.6147)
CTSB | 227961_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1432)
CTSB | 227961_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2382)
CTSB | 227961_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5819)
CTSB | 227961_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1358)
CTSB | 227961_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSB | 227961_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6025
p=0.7622
p=0.762
p=0.5115
p=0.5569
p=0.4332
p=0.3672
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2